

ZACHARY SMITH

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PROFESSIONAL SUMMARY

Data Science graduate from Pennsylvania State University with hands-on experience in healthcare analytics, machine learning, and data visualization. Proficient in Python, SQL, React, Next.js, and modern AI/ML frameworks. Skilled at transforming complex telemetry and operational datasets into automated reporting pipelines, predictive models, and high-impact analytics solutions.

EDUCATION

B.S. Data Science – Pennsylvania State University, University Park, PA | 2022 – May 2026

High School Diploma – Mira Costa High School, Manhattan Beach, CA | 2018 – 2022

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, JavaScript (ES6+), HTML5/CSS3
- **AI & ML:** scikit-learn, XGBoost, TensorFlow, PyTorch, HuggingFace Transformers, FAISS, VADER NLP
- **Web & Frameworks:** React, Next.js, Tailwind CSS, Recharts, Chart.js, Node.js
- **Data & Tools:** MySQL, SSMS, Tableau, Git/GitHub, VSCode, Pandas, NumPy, Matplotlib

PROFESSIONAL EXPERIENCE

Data Science Intern – Pacific Healthworks | February 2023 – July 2026

- Structured, analyzed, and visualized healthcare datasets to identify key operational trends and actionable clinical insights.
- Leveraged MySQL, SSMS, and Tableau for comprehensive data cleaning, transformation, and interactive dashboard modeling.
- Presented data-driven findings and strategic recommendations to clinical leadership to optimize decision-making.
- Automated workflow processes and data pipelines to generate updated reporting tables and executive analytics.

Intern – Career Movement | May 2023 – Aug 2023

- Conducted market research and sentiment analysis on employment trends to inform talent placement strategies.
- Performed structured data entry and database maintenance to ensure cross-functional data integrity.

Concession Stand Worker – Pennsylvania State University | 2024 – 2025

- Collaborated within high-volume team environments to deliver efficient customer service and operational inventory tracking.

FEATURED DATA SCIENCE PROJECTS

Game Recommendation LLM Engine | *Python, HuggingFace, Transformers, SQL, FAISS*

- Built a semantic recommendation engine using HuggingFace transformer embeddings to match user preferences with catalog descriptions.
- Designed a high-performance FAISS vector index enabling sub-millisecond similarity searches at scale.

Steam Game Market Success Predictor | *Python, SteamSpy API, XGBoost, VADER NLP, Matplotlib*

- Developed an end-to-end ML classification pipeline achieving 85.7% accuracy predicting product market success.
- Engineered features from live telemetry (concurrent users, pricing) and processed customer reviews via VADER sentiment analysis.

Player Churn & Survival Analytics Model | *Python, XGBoost, Scikit-Learn, Cox Hazards, Pandas*

- Engineered a hybrid survival analysis & classification pipeline to predict player churn velocity from engagement telemetry.

DataPulse & BetTruth Analytics Portals | *Next.js 16, React, TypeScript, Chart.js, Monte Carlo Sim*

- Built full-stack interactive analytics dashboards for Monte Carlo risk simulations, executive KPIs, and retention heatmaps.